

TCC805x MCU BSP

API Specification for Input Capture Time Counter

Rev. 1.00 [A]

2020-11-12

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1 INTRODUCTION

This document describes the input capture time counter device driver (hereinafter referred to as ICTC driver) of TCC805x MCU BSP. For more detailed information of the input capture time counter in the TCC805x MCU Sub-system, refer to Chapter 15 Input Capture Time Counter (ICTC) of the *“PART 11-MICOM SUB-SYSTEM”* in the *“TCC805x Full Specification”*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

ICTC	Input Capture Time Counter
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3 SOURCE FILES

3.1 Location of Source Files

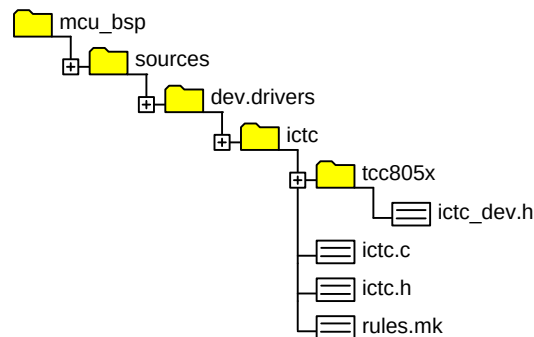


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- ictc_dev.c
 - ICTC specific header by chipset.
- ictc.c
 - ICTC common source code
- ictc.h
 - ICTC common header including API & DEFINE for user
- rules.mk
 - Make rules

4 CONSTANTS

4.1 ICTC Channel Index

Constant values for the ICTC channel index

Declaration

```
#define ICTC_CH_0 (0UL)
#define ICTC_CH_1 (1UL)
#define ICTC_CH_2 (2UL)
```

Description

<i>Value</i>	<i>Description</i>
<i>ICTC_CH_0</i>	<i>Index of ICTC channel0</i>
<i>ICTC_CH_1</i>	<i>Index of ICTC channel1</i>
<i>ICTC_CH_2</i>	<i>Index of ICTC channel2</i>

Remark

None

4.2 Channel Information Value

Constant values for ICTC channel information value

Declaration

```
#define ICTC_SUB_ICTC_ENABLED
#define ICTC_TOTAL_CHANNEL_NUM      (3UL)
#define ICTC_DIFF_MARGIN            (0x800UL)

#define ICTC_TIMESTAMP_RECORD_ENABLE
#define ICTC_TIMESTAMP_RECORD_MAXCNT (15UL)
```

Description

Value	Description
<i>ICTC_SUB_ICTC_ENABLED</i>	<i>Enable Redundancy ICTC for safety feature</i>
<i>ICTC_TOTAL_CHANNE_NUM</i>	<i>Total count of the ICTC channel</i>
<i>ICTC_DIFF_MARGIN</i>	<i>Margin Value between Main ICTC and Sub-ICTC for safety feature</i>
<i>ICTC_TIMESTAMP_RECORD_ENABLE</i>	<i>Enable Time-Stamp Record</i>
<i>ICTC_TIMESTAMP_RECORD_MAXCNT</i>	<i>Maximum count of Record Time-stamp</i>

Remark

None

5 STRUCTURES

5.1 ICTC callBack_t

Structure to register a callback function that notifies a user of ICTC interrupt handle completion

Declaration

```
typedef struct _ICTCAllBack_  
{  
    int32          cbChannel;  
    ICTCAllBack    cbCallBackFunc;  
} ICTCAllBack_t;
```

Description

Value	Description
<i>uiChannel</i>	<i>The ICTC channel Number 0 ~ 2</i>
<i>ucCallBackFunc</i>	<i>The pointer of callback function. Refer to the ICTCAllBack.</i>

Remark

None

6 CALLBACK FUNCTIONS

6.1 ICTCCallback

When the ICTC input interrupt handle is done, the ICTC driver calls this function to deliver the calculated period and duty value to a user.

Declaration

```
typedef void (*ICTCCallback)
(
    uint32 uiChannel,
    uint32 uiPeriod,
    uint32 uiDuty
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>ICTC channel Number 0 ~ 2</i>
<i>uiPeriod</i>	<i>Period value of input signal. This value is delivered to user.</i>
<i>uiDuty</i>	<i>Duty value of input signal. This value is delivered to user.</i>

Return

None

Remark

None

7 APIs

7.1 ICTC_Init

Enables peripheral clock for the ICTC module.

Declaration

```
void ICTC_Init  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.2 ICTC_Deinit

Disables peripheral clock for the ICTC module.

Declaration

```
void ICTC_Deinit  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

7.3 ICTC_EnableCapture

Starts the input signal capture.

Declaration

```
void ICTC_EnableCapture
(
    uint32    uiChannel
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>The ICTC channel Number 0 ~ 2</i>

Return

None

Remark

None

7.4 ICTC_DisableCapture

Stops the input signal capture.

Declaration

```
void ICTC_DisableCapture
(
    uint32    uiChannel
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>The ICTC channel Number 0 ~ 2</i>

Return

None

Remark

None

7.5 ICTC_SetOpEnCtrlCounter

Sets the ICTC counter types in *OP_EN_CTRL* register.

Declaration

```
void ICTC_SetOpEnCtrlCounter
(
    uint32    uiChannel,
    uint32    uiCntEnValue
);
```

Parameters

Value	Description
<i>uiChannel</i>	The ICTC channel Number 0 ~ 2
<i>uiCntEnValue</i>	Counter type value which is defined in <i>ictc_dev.h</i> (<i>ICTC_OPEN_CTRL_XXXX_EN</i>)

Return

None

Remark

There are 6 counter types.

- Pre-scaler counter (TCLK_EN)
- Time stamp counter (TSCNT_EN)
- Filter counter (FLTCNT_EN)
- Time-out counter (TOCNT_EN)
- Edge counter (ECNT_EN)
- Period/duty compare counter (PDCNT_EN)

7.6 ICTC_SetOpModeCtrlReg

Sets the ICTC mode and input port (*F_IN_SEL*) in *OP_MODE_CTRL* register.

Declaration

```
sint32 ICTC_SetOpModeCtrlReg  
(  
    uint32 uiChannel,  
    uint32 uiModeValue  
);
```

Parameters

Value	Description
<i>uiChannel</i>	The ICTC channel Number 0 ~ 2
<i>uiCntEnValue</i>	Mode value which is defined in <i>ictc_dev.h</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	Success
<i>SAL_RET_FAILED</i>	Failure

Remark

Refer to “TCC805x Full Specification” for MODE information.

7.7 ICTC_SetIRQCtrlReg

Enables the ICTC interrupt. The ICTC_SetIRQCtrlReg sets selected IRQ types in **IRQ_CTRL** register (IRQ_EN[31:24]).

Declaration

```
void ICTC_SetIRQCtrlReg
(
    uint32 uiChannel,
    uint32 uiIRQValue
);
```

Parameters

Value	Description
uiChannel	The ICTC channel Number 0 ~ 2
uiIRQValue	IRQ type value

Return

None

Remark

Refer to “TCC805x Full Specification” to get **IRQ_EN** information.

7.8 ICTC_SetCallbackFunc

Registers the ICTC callback function.

Declaration

```
void ICTC_SetCallbackFunc
(
    uint32      uiChannel,
    ICTCCallback pCallbackFunc
);
```

Parameters

Value	Description
uiChannel	The ICTC channel Number 0 ~ 2
pCallbackFunc	The pointer of callback function. Refer to the ICTCCallback .

Return

None

Remark

None

7.9 ICTC_SetTimeoutValue

Sets the ICTC timeout value. If input signal is slower than ICTC clock, for example, ICTC: 100MHz / input: 5MHz—the timeout count increases. Then a timeout interrupt is generated when the timeout count reaches timeout value.

Declaration

```
void ICTC_SetTimeoutValue
(
    uint32 uiChannel,
    uint32 uiTimeoutValue
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in]</i> ICTC channel Number 0 ~ 2
<i>uiTimeoutValue</i>	<i>[in]</i> ICTC Timeout value

Return

None

Remark

None

7.10 ICTC_SetEdgeMatchingValue

Sets the ICTC edge counter matching values. The ICTC edge counter matching value means ignored input signal counter matching value to treat noise from starting capture.

Declaration

```
void ICTC_SetEdgeMatchingValue
(
    uint32 uiChannel,
    uint32 uiRisingEdgeMat,
    uint32 uiFallingEdgeMat,
    uint32 uiEdgeCounterMat
);
```

Parameters

Value	Description
uiChannel	The ICTC Channel value 0 ~ 2
uiRisingEdgeMat	Rising edge matching value
uiFallingEdgeMat	Falling edge matching value
uiEdgeCounterMat	Edge counter matching value

Return

None

Remark

None

7.11 ICTC_SetCompareRoundValue

Sets the ICTC period and duty compare round values (*PRD_CMP_RND*, *BI*). The *PRD_CMP_RND* means frequency counter masking value to compare previous frequency and current frequency. If previous frequency counter value and current frequency counter value are not the same, the ICTC interrupt occurs. The *DUTY_CMP_RND* is similar to *PRD_CMP_RND*

Declaration

```
void ICTC_SetCompareRoundValue
(
    uint32 uiChannel,
    uint32 uiPeriodCompareRound,
    uint32 uiDutyCompareRound
);
```

Parameters

Value	Description
<i>uiChannel</i>	<i>[in]</i> ICTC Channel value 0 ~ 2
<i>uiPeriodCompareRound</i>	<i>[in]</i> Period compare round value
<i>uiDutyCompareRound</i>	<i>[in]</i> Duty compare round value

Return

None

Remark

None

8 HOW TO USE ICTC DRIVER

You should initialize and configure the ICTC module before starting input capture. Refer to *ictc_test.c* file to get example code.

8.1 Operation of ICTC

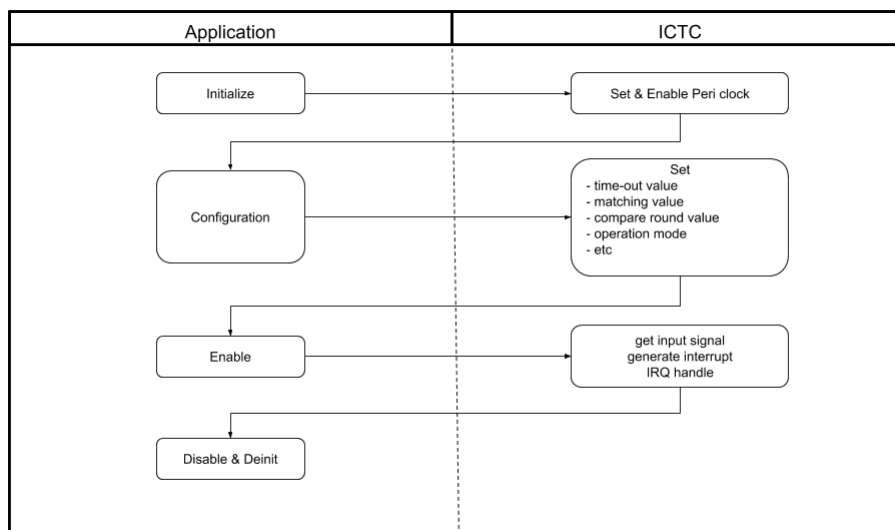


Figure 8.1 Diagram for ICTC driver

8.2 Example for using ICTC Driver

```

{
  //Initialize
  ICTC_Init();

  //Configuration
  ICTC_TestSetGpio();

  ICTC_SetCallBackFunc(...); //optional
  ICTC_SetTimeoutValue(...); //optional
  ICTC_SetEdgeMatchingValue(...); //optional
  ICTC_SetCompareRoundValue(...); //optional

  ICTC_SetOpModeCtrlReg(...);
  ICTC_SetOpEnCtrlCounter(...);
  ICTC_SetIRQCtrlReg(...);

  // Enable (Start input signal capture)
  ICTC_EnableCapture();

  // ...
  // process IRQ generated from duty & period change of input signal

  // Disable (Stop input signal capture)
  ICTC_DisableCapture();
}
  
```

9 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

10 REVISION HISTORY

Rev. 1.00: 2020-11-12

- First version release

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